SE Lab 3: Architectural Style Justification

Telemedicine Slot Booking & Prescription Portal

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SECTION: 5B

Architecture Selection: We chose Microservices Architecture for the Telemedicine Slot Booking & Prescription Portal.

Two Scenario-Related Reasons

- **Independent, uneven scaling needs:** The Consultation Service experiences sharp load spikes during clinic hours as patients join video calls (FR-003), while the Prescription Service and Booking Service see steadier, lighter traffic. Microservices let each service scale on its own hardware/instances instead of over-provisioning the entire system, which directly supports NFR-001 (available slots must display within 5 seconds even under load).
- **Fault isolation across critical vs. non-critical functions:** Slot booking (FR-001, FR-002) and live consultation (FR-003) are high-priority, always-on functions, whereas administrative availability management (FR-005) is lower priority. Splitting these into separate services means a failure or slow deployment in one service (e.g., the Prescription Service cannot take down booking or in-progress consultations, unlike a monolithic or tightly layered design where a single fault can cascade.

Security Advantage

Each microservice enforces its own authentication/authorization boundary by using the dedicated Authentication Service before performing sensitive operations.. In particular, the Prescription Service and Consultation Service independently verify that the requesting patient is the authenticated owner of that appointment before releasing data or granting consultation access. This directly satisfies NFR-002 (only the authenticated patient and their assigned doctor may access a given prescription) and limits the blast radius of a compromised service, since no single service holds unchecked access to another service's data store.

Performance Benefit

Because the Booking Service, Consultation Service, and Prescription Service are deployed and scaled independently, a surge in patients searching for and booking slots (FR-001, FR-002) does not compete for the same compute resources as active video consultations or prescription downloads. This isolation keeps slot-search response times consistently low, supporting the NFR-001 requirement that available slots be displayed within 5 seconds in at least 19 of 20 consecutive searches, even while other parts of the system are under heavy simultaneous use.

Component Diagram:

